

# MUHAMMED ALTHAF

AI ENGINEER • CLOUD ARCHITECT • SOLUTION ARCHITECT • SOFTWARE DEVELOPMENT LEAD

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## PROFESSIONAL PROFILE

AI-focused Cloud and Solution Architect with 7+ years of experience designing and delivering scalable products across healthcare, fintech, satellite telecommunications, vehicle telematics, legal data, IoT, and enterprise systems. Leads AI-first delivery using LLM APIs, RAG, autonomous agents, hybrid retrieval, and multimodal analysis, backed by hands-on expertise in AWS, Azure, Google Cloud, microservices, event-driven systems, web, mobile, and DevOps. Combines architecture ownership with full-stack execution, technical leadership, cloud migration, integrations, and production delivery.

## CORE EXPERTISE

|                       |                                                                                                             |
|-----------------------|-------------------------------------------------------------------------------------------------------------|
| AI Engineering        | LLM applications, RAG, agents, prompt engineering, embeddings, hybrid search, multimodal and video analysis |
| Cloud Architecture    | AWS, Microsoft Azure, Google Cloud, migrations, containers, CI/CD, Linux, Nginx, observability              |
| Solution Architecture | Microservices, events, APIs, SSO, data pipelines, IoT, fintech, and geospatial systems                      |
| Software Development  | Python, C#, JavaScript/TypeScript, PHP, React, Angular, Vue, .NET, Django, FastAPI, Laravel, Flutter        |

## TECHNICAL SKILLS

**AI, LLM & Agents:** OpenAI API, Claude API, Gemini, Google Vertex AI, Perplexity, LangChain, LangGraph, CrewAI, AutoGen, Hermes Agent, Pi Agent, agent orchestration, agent swarming, tool calling, prompt engineering, LLM fine-tuning, Claude Code, Cursor, harness engineering

**RAG, Search & Multimodal:** RAG architecture, embeddings, vector databases, Qdrant, Pinecone, ChromaDB, BM25, hybrid retrieval, knowledge systems, web search, AI video analysis, image analysis, NLP, OCR, PyTesseract, classification, entity extraction

**Cloud & DevOps:** AWS, Azure, Google Cloud, Docker, Kubernetes, Azure DevOps, GitHub Actions, Jenkins, CI/CD, Linux, Nginx, Coolify, server management, cloud migration

**Architecture & Integration:** Solution architecture, system design, microservices, event-driven architecture, REST APIs, OAuth 2.0, JWT, Keycloak SSO, Kafka, RabbitMQ, Celery, Apache Flink, Node-RED, blockchain/Ganache, IoT, real-time systems

**Application Development:** Python, JavaScript, TypeScript, C#, PHP, Dart, React.js, Next.js, Angular, Vue.js, Node.js, Express.js, Django, FastAPI, .NET Core, ASP.NET, Laravel, Flutter, React Native

**Data & Platforms:** PostgreSQL, MySQL, SQL Server, MongoDB, DynamoDB, Redis, Firebase, Xero, Stripe, PayPal, Telr, Leaflet, OpenStreetMap, Google Maps API, SendGrid

## PROFESSIONAL EXPERIENCE

### Project Manager / Full Stack Developer

Mar 2025–Present

Digital-01 | Remote

- Lead multimodal AI products for meeting intelligence, including uploaded or Loom-hosted recording, screen-content, and transcript analysis with conversational question answering, plus reel analysis and generation.
- Drive modernization of a large Drupal 7 estate through AI-assisted development, server management, and Azure cloud migration.
- Build automated company website generation and a Payload CMS architecture for high-content sites.
- Design an OpenClaw cricket-data parser with LLM Q&A; a repository-aware Slack bot with approval-gated GitHub inspection and ticket workflows, and end-to-end app delivery through specialized Hermes agents.

## Software Developer

May 2023–Feb 2025

S2t Cyberspace | Remote

- Owned architecture, development, and deployment of a feature-rich healthcare platform using a micro-frontend approach.
- Designed secure Keycloak SSO and OpenAI-powered analytics for personalized treatment-plan insights.
- Delivered payment-processor and Xero integrations, secure real-time chat, and location-based healthcare capabilities.

## Co-Founder & Technical Lead

Jan 2023–Apr 2023

Bitroxy Solutions | Dubai, UAE

- Owned technical roadmap, architecture decisions, delivery standards, and feature prioritization for a chef marketplace.
- Built an IoT and blockchain proof of concept linking safety sensors through Node-RED to a local Ganache blockchain.
- Integrated Telr payments and established coding standards, automated testing, and CI/CD workflows.

## Software Developer

Jun 2022–Jan 2023

Esoftsat (GlobalTT) | Dubai, UAE

- Enhanced an enterprise BSS/OSS platform supporting satellite telecommunications providers across multiple regions.
- Worked across NMS operations, bandwidth allocation, channel control, NX Allot, and satellite-equipment integration.
- Built ERP monitoring capabilities and optimized SQL Server queries for reliability and performance.

## Senior Software Developer

Oct 2019–May 2022

V Zone International LLC | Dubai, UAE

- Delivered an end-to-end vehicle-tracking platform covering IoT ingestion, real-time tracking, geofencing, alerts, and fleet reporting.
- Designed fault-tolerant Kafka data pipelines and proposed an Apache Flink streaming architecture for cloud migration.
- Created device-management tools and an offline-first, cross-platform Flutter application.

## Software Developer

Jul 2018–Jul 2019

UniCourt | Mangalore, India

- Designed distributed scraping and microservices architecture to process millions of court records across jurisdictions.
- Developed ML models for classification and entity extraction, with custom PyTesseract OCR and preprocessing.
- Implemented event-driven communication patterns for scalable legal-data processing pipelines.

## SELECTED ARCHITECTURE & PRODUCT PORTFOLIO

**Meeting & Video Intelligence** — Analyzes uploaded or Loom-hosted meetings, screen activity, and transcripts for contextual Q&A; includes reel analysis and generation.

*Technology: Multimodal LLMs, video analysis, transcription, RAG, conversational search*

**Agentic Engineering Platform** — Repository-aware Slack assistance with approval-gated GitHub inspection, demo execution, technical Q&A, ticket workflows, and end-to-end delivery through specialized Hermes agents.

*Technology: Hermes Agent, Slack, GitHub, multi-agent orchestration, tools, approval gates*

**Generative Web & Content Automation** — Generates tailored company websites and supports content-heavy publishing through Payload CMS.

*Technology: LLMs, automated website generation, Payload CMS, full-stack delivery*

**Cricket Data Intelligence** — Uses OpenClaw to structure unstructured cricket data in a database and answer natural-language questions through a connected LLM.

*Technology: OpenClaw, structured extraction, databases, LLM question answering*

**Healthcare Platform** — Secure care platform with personalized AI analytics, fintech integrations, real-time chat, and geospatial services.

*Technology: Micro-frontends, Keycloak, OpenAI, Xero, Leaflet/OpenStreetMap*

**Vehicle Tracking & Fleet Intelligence** — IoT telematics platform with real-time tracking, fault-tolerant ingestion, reporting, and offline mobile support.

*Technology: Kafka, Apache Flink architecture, Flutter, IoT*

**Legal Data Intelligence** — Distributed acquisition and AI processing of millions of court records with OCR, classification, and entity extraction.

*Technology: Python, PyTesseract, ML, microservices, event-driven processing*